GHANA COMMUNICATION TECHNOLOGY UNIVERSITY

FACULTY OF COMPUTING AND INFORMATION SYSTEMS

DEPARTMENT OF COMPUTER SCIENCE

BSc./Dip. (Computer Science, Cyber Security, Data Science & Analytics, Software Engineering)
(Level 100 MORNING)

END-OF-FIRST-SEMESTER EXAMINATION, APRIL 2025

MATH 105: LINEAR ALGEBRA

Duration: 2 hrs 15 mins

INDEX NUMBER: ....

Instructions to students:

1. Be sure to fill in your index number on both the question paper and answer booklet before starting the examination.
2. Answer ***all questions*** in SECTION A and ***only two questions*** in SECTION B.
3. Read all questions and possible answers *carefully*.
4. For multiple-choice questions, circle ***only one*** letter indicating your answer and write the letter in the answer booklet. Ambiguous selections will be marked wrong!
5. Multiple-choice questions in SECTION A must be answered on the question paper and in the answer booklet while questions in SECTION B must be answered in the answer booklet.
6. Submit the question paper together with the answer booklet.

SECTION A (30 Marks: 1 mark each)

ANSWER ALL QUESTIONS IN THIS SECTION

Write the letter for the correct answer in the answer booklet and circle it on the question paper

Use the system of linear equations below to answer questions 1 – 4.

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

1. Which of the following best describes the coefficient matrix of the system of linear equations above?

   A) $\begin{bmatrix} -1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$

   B) $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$

   C) $\begin{bmatrix} 0 & -2 & 1 \\ 1 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$

   D) $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$

2. Which of the following best describes the augmented matrix of the system of linear equations above?

   A) $\begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix}$

   B) $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$

   C) $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix}$

   D) $\begin{bmatrix} 0 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 0 & -5 & 10 \end{bmatrix}$

3. What is the size of the coefficient matrix of the system of linear equations above?

   A) 2×3
   B) 3×2
   C) 3×3
   D) 4×4

4. Find the solution set of the system of linear equations above.

   A) (1,0,1)
   B) (1,0,-1)
   C) (0,1,-2)
   D) (1,1,0)

5. Which of the following augmented matrices is in reduced row echelon form?

A) $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

B) $\begin{bmatrix} 1 & 0 & 4 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

C) $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

D) $\begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

6. Let the vectors $u_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $u_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$, and $y = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$. Determine the weights, x_1 and x_2 that would make the linear combination $x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$ valid.

A) $x_1=2,\ x_2 = 3$
B) $x_1=3,\ x_2 = 2$
C) $x_1=-2,\ x_2 = -3$
D) $x_1=2,\ x_2 = -3$

7. Find values of x and y, if any, that will make the matrices below equal.

$$\begin{bmatrix} x+y & 3+x \\ -3 & 3y \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -3 & 6 \end{bmatrix}$$

A) $x = 1, y = 2$
B) $x = -1, y = 2$
C) $x = -1, y = -2$
D) $x = 2, y = 1$

8. Find values of $x - y$, if any, that will make the matrices below equal.

$$\begin{bmatrix} x+y & 3+x \\ -3 & 3y \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -3 & 6 \end{bmatrix}$$

A) 1
B) -1
C) 2
D) -2

9. Find the element at the second row second column of the product of the matrices below.

$$\begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}$$

A) -1
B) 0
C) 13
D) 21

10. If the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ x-3 & -2 & -1 \\ x-2 & -1 & -5 \end{bmatrix}$ is symmetric, find the value of x.

A) 0
B) 3
C) 5
D) 6

11. Given that matrix $A = \begin{bmatrix} 16 & 1 \\ -1 & 2 \end{bmatrix}$, find the element of the inverse, $A^{-1}{}_{(22)}$.

A) $-\frac{1}{33}$
B) $\frac{1}{33}$
C) $\frac{2}{33}$
D) $\frac{16}{33}$

12. Given that matrix $P = \begin{bmatrix} 4 & 1 & -1 \\ 4 & 4 & -1 \\ -1 & -1 & 1 \end{bmatrix}$, find the element of the inverse, $P^{-1}{}_{(31)}$.

A) $-\frac{1}{3}$
B) 0
C) $\frac{1}{3}$
D) $\frac{4}{3}$

13. Given that matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find the element of the minor, $Min(A)_{(21)}$.

A) 1
B) 2
C) 3
D) 4

14. Given that matrix $P = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 5 \\ -1 & 2 & 3 \end{bmatrix}$, find the element of the cofactor, $Cof(P)_{(23)}$.

A) 0
B) 3
C) 6
D) 7

15. Find the solution of the variable z for the system of linear equations below.

$$2x + y - 3z - 8 = 0$$
$$x + 3y + 2z - 5 = 0$$
$$3x + 2y - z - 6 = 0$$

A) $-\frac{31}{14}$

B) $-\frac{15}{14}$
C) $-\frac{49}{14}$
D) $-\frac{18}{14}$

16. Determine the rank of the matrix $Q = \begin{bmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$.

A) 1
B) 2
C) 3
D) Not defined

17. Determine the rank of the matrix $M = \begin{bmatrix} 1 & 2 & 8 \\ 4 & 7 & 6 \\ 9 & 5 & 3 \end{bmatrix}$.

A) 1
B) 2
C) 3
D) Not defined

18. Given $\begin{vmatrix} -6 & 2 \\ 8 & x \end{vmatrix} = 20$, find the value of x^2.

A) -6
B) 6
C) 12
D) 36

19. The determinant of the matrix $X = \begin{bmatrix} 5 & 0 & 2 \\ 4 & 2 & 4 \\ 6 & 3 & 6 \end{bmatrix}$ is given as...

A) $|X| = 0$
B) $|X| = 24$
C) $|X| = 60$
D) $|X| = 84$

20. Determine the dimension of A^T given that matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & -2 \end{bmatrix}$.

A) 2×3
B) 2×1
C) 3×2
D) 1×2

21. Describe the nature of matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ using its dimension.

A) Row matrix
B) Column matrix
C) Square matrix
D) Rectangular matrix

22. Given that the dimensions of matrices A and B are $(m \times n)$ and $(n \times p)$ respectively. What will be the dimension of matrix AB, the product of A and B?

A) $(m \times n)$
B) $(n \times p)$
C) $(m \times p)$
D) Not defined

23. Find the magnitude of the vector joining the points $A(2,\ 1)$ and $B(8,\ 9)$.

A) 10
B) 36
C) 64
D) 100

24. Find the position vector of the point $(-1,0,2)$ in the R^3 plane.

A) $-i + 2k$
B) $-i + 2j$
C) $2j - k$
D) $-i + 2j + k$

25. Find the eigenvalues of the matrix $M = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}$.

A) 1 and 2
B) -1 and 2
C) 1 and -2
D) -1 and -2

26. Which of the following is a difference between the eigenvalues of the matrix?
$P = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}$?

A) -1
B) 0
C) 2
D) 3

27. Given that the vectors u = (-14, 4), v = (5, -2) and w = (6, -4). Express the vector w, as a linear combination of vectors u and v.

A) $w = u - 4v$
B) $w = -u + 4v$
C) $w = u + 4v$
D) $w = -u - 4v$

28. Given that $\lambda = \begin{bmatrix} 4/3 \\ 1 \end{bmatrix}$ is an eigenvector of the matrix $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$, find the eigenvalue associated with the eigenvector.

A) -5
B) -2
C) 2
D) 5

29. Convert the position vector $\boldsymbol{a} = 3i - 4j$ to the magnitude bearing form.

A) $(5,\ 53°)$
B) $(25,\ 53°)$
C) $(5,\ 143°)$
D) $(25,\ 143°)$

30. Determine the scalar product of the two vectors **a** = (−1,0,1) and **b** = (0, 2, 3).

A) 3
B) 6
C) 7
D) 8

End of Section A

SECTION B (30 Marks)

Answer TWO questions: Question 31 and any other question

Question 31 (COMPULSORY)

Use the system of linear equations below to answer the following questions.

$$2x + y + 4z - 12 = 0$$
$$4x + 11y - z - 33 = 0$$
$$8x - 3y + 2z - 20 = 0$$

a. Write the standard form $AX = B$ of the system of linear equations and find the augmented matrix. **[4 marks]**

b. Find the inverse of the coefficient matrix A, A^{-1} of the system of linear equations using the adjoint of A. **[6 marks]**

c. Find the solution of the system of linear equations above using the Cramer's rule approach. **[10 marks]**

Question 32

a. Find the cross product a × b of the vectors **a** = i − j + 2k and **b** = 2i + k. **[5 marks]**

b. Given the vectors u=(2, 4), v=(1, -2) and w=(6, -4). Express the vector *w*, as a linear combination of u and v. **[5 marks]**

Question 33

A matrix is given as $A = \begin{bmatrix} 3 & 10 \\ 2 & 4 \end{bmatrix}$ such that $A\boldsymbol{x} = \lambda\,\boldsymbol{x}$, where λ and $\boldsymbol{x}$ are the eigenvalues and eigenvectors vectors respectively.

a. Find the characteristic equation. **[2 marks]**

b. Determine the associated eigenvalues. **[2 marks]**

c. Find the corresponding eigenvectors. **[6 marks]**

BEST WISHES

Examiner: Opoku Gyabaah